Chapter 51

USB Serial/JTAG Controller (USB_SERIAL_JTAG)

ESP32-P4 contains a USB Serial/JTAG Controller. This unit can be used to program the SoC's flash, read program output, as well as attach a debugger to the running program. All of these are possible for any computer with a USB host (hereafter referred to as 'host') without any active external components.

51.1 Overview

While programming and debugging an ESP32-P4 project using the UART and JTAG functionality is certainly possible, it has a few downsides. First of all, both UART and JTAG take up IO pins and as such, fewer pins are left usable for controlling external signals in software. Additionally, an external chip or adapter is needed for both UART and JTAG to interface with a host computer, which means it will be necessary to integrate these two functionalities in the form of external chips or debugging adapters.

In order to alleviate these issues, ESP32-P4 provides a USB Serial/JTAG Controller, which integrates the functionality of both a USB-to-serial converter as well as a USB-to-JTAG adapter. As this device directly interfaces with an external USB host using only the two data lines required by USB 2.0, only two pins are required to be dedicated to this functionality for debugging ESP32-P4.

51.2 Features

The USB Serial/JTAG controller has the following features:

- USB Full-speed device; Hardwired for CDC-ACM (Communication Device Class - Abstract Control Model) and JTAG adapter functionality
- CDC-ACM:
  - Integrates CDC-ACM adherent serial port emulation (plug-and-play on most modern OSes)
  - Supports host controllable chip reset and entry into download mode
- JTAG adapter functionality:
  - Allows fast communication with CPU debugging core using a compact representation of JTAG instructions
- A control endpoint, a dummy interrupt endpoint, two bulk input endpoints, and two bulk output endpoints; Up to 64-byte data payload size
- Internal PHY: very few or no external components needed to connect to a host computer

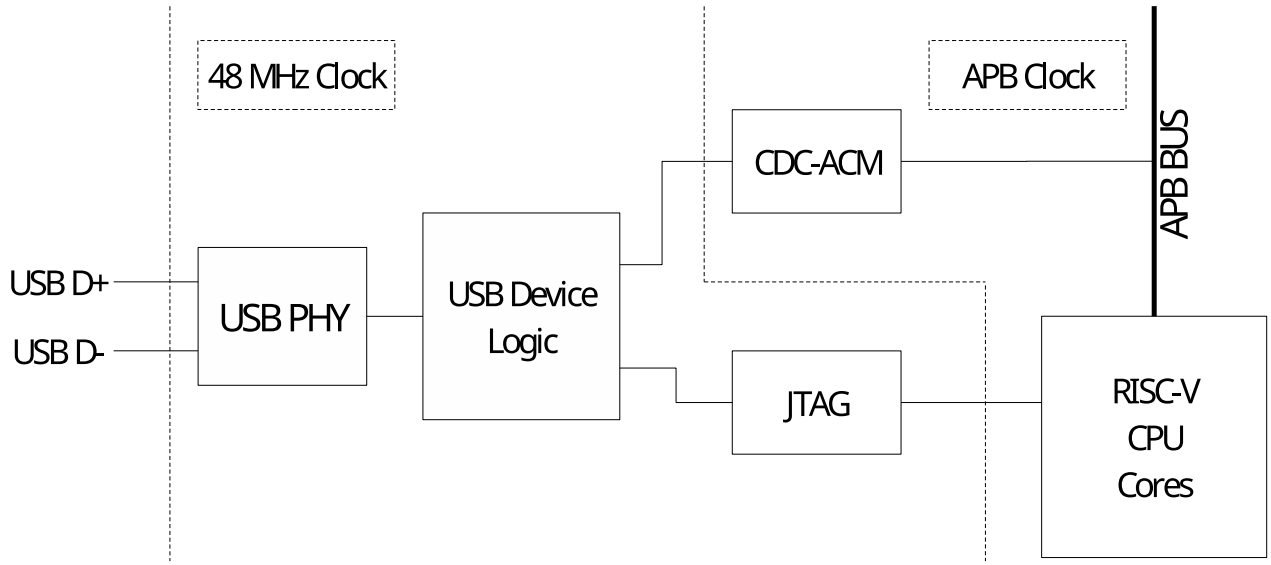


Figure 51.2-1. USB Serial/JTAG High Level Diagram

As shown in Figure 51.2-1, the USB Serial/JTAG controller consists of a USB PHY, a USB device interface, a JTAG command processor, a response capture unit, and the CDC-ACM registers. The PHY and device interface are clocked from a 48 MHz clock derived from the baseband PLL (BBPLL); the software-accessible side of the CDC-ACM block is clocked from APB_CLK. The JTAG command processor is connected to the JTAG debugging unit of the main processor; the CDC-ACM registers are connected to the APB bus and as such can be read from and written to by software running on the main CPU.

Note that while the USB Serial/JTAG device supports USB 2.0 standard, it only supports Full-speed (12 Mbps) mode but not other modes that the USB 2.0 standard introduced, e.g., the High-speed (480 Mbps) mode.

Figure 51.2-2 shows the internal details of the USB Serial/JTAG controller on the USB side. The USB Serial/JTAG controller consists of a USB 2.0 Full-speed device. It contains a control endpoint, a dummy interrupt endpoint, two bulk input endpoints, and two bulk output endpoints. Together, these form a USB composite device, which consists of a CDC-ACM USB class device as well as a vendor-specific device implementing the JTAG interface. On the SoC side, the JTAG interface is directly connected to the RISC-V CPU's debugging interface, allowing debugging of programs running on that core. Meanwhile, the CDC-ACM device is exposed as a set of registers, allowing a program on the CPU to read and write from it. Additionally, the ROM startup code of the SoC contains code that allows the user to reprogram attached flash memory using this interface.

Note:
To get specific information about what endpoint numbers are used for what functionality, please parse the relevant USB descriptors as returned by the device. This guarantees compatibility with other devices with the same functionality but a different implementation.

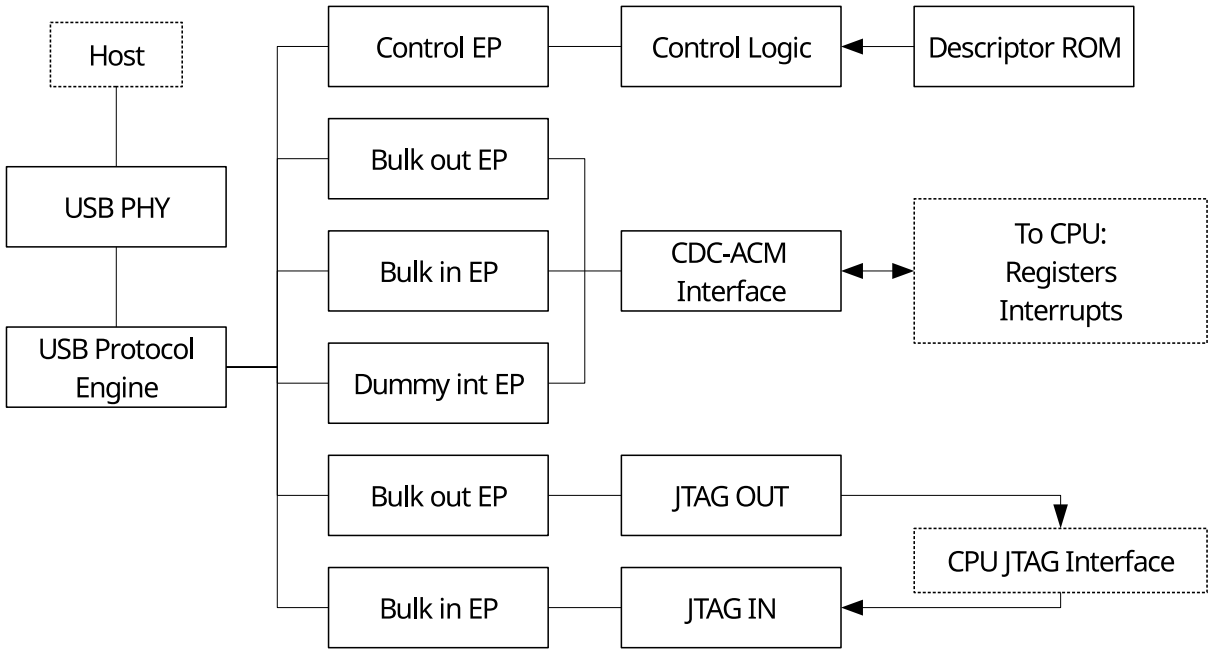


Figure 51.2-2. USB Serial/JTAG Block Diagram

51.3 Functional Description

The USB Serial/JTAG controller interfaces with a USB host processor on one side, and with the CPU debugging hardware as well as the software that communicates through the CDC-ACM port on the other side.

51.3.1 CDC-ACM USB Interface Functional Description

The CDC-ACM interface adheres to the standard USB CDC-ACM class for serial port emulation. It contains a dummy interrupt endpoint (which will never send any events, as they are not implemented nor needed) and a Bulk IN as well as a Bulk OUT endpoint for the host's received and sent serial data respectively. These endpoints can handle 64-bytes packet at a time, allowing high throughput. As CDC-ACM is a standard USB device class, a host generally can function without any special installation procedures. That is to say, when the USB debugging device is properly connected to a host, the operating system should show a new serial port moments later.

The CDC-ACM interface accepts the following standard CDC-ACM control requests:

Table 51.3-1. Standard CDC-ACM Control Requests

Command	Action
SEND_BREAK	Accepted but ignored (dummy)
SET_LINE_CODING	Accepted, value sent is readable in software
GET_LINE_CODING	By default, returns 9600 baud, no parity, 8 data bits, 1 stop bit (Can be changed through software)
SET_CONTROL_LINE_STATE	Set the state of the RTS/DTR lines. See Table 51.3-2

Aside from general-purpose communication, the CDC-ACM interface can also be used to reset ESP32-P4 and optionally make it enter download mode to flash new firmware. This can be realized by setting the RTS and DTR lines on the virtual serial port.

Table 51.3-2. CDC-ACM Settings with RTS and DTR

RTS	DTR	Action
0	0	Clear download mode flag
0	1	Set download mode flag
1	0	Reset ESP32-P4
1	1	No action

Note that if the download mode flag is set when ESP32-P4 is reset, ESP32-P4 will reboot into download mode. When this flag is cleared and the chip is reset, ESP32-P4 will boot from flash. For specific sequences, please refer to Section 51.4. All these functions can also be disabled by programming various eFuses. Please refer to Chapter 8 *eFuse Controller (EFUSE)* for more details.

51.3.2 CDC-ACM Firmware Interface Functional Description

The CPU can interact with the USB Serial/JTAG controller as the module is connected to the internal APB bus of ESP32-P4. This is mainly used to read and write data from and to the virtual serial port on the attached host.

USB CDC-ACM serial data is sent to and received from the host in packets of 0 to 64 bytes in size. When enough CDC-ACM data has accumulated in the host, the host sends a packet to the CDC-ACM receive endpoint, and the USB Serial/JTAG controller accepts this packet if it has a free buffer. Conversely, the host checks periodically if the USB Serial/JTAG controller has a packet ready to be sent to the host, and if so, receives this packet.

Firmware can get notified of new data from the host in one of the following two ways. First of all, the USB_SERIAL_JTAG_SERIAL_OUT_EP_DATA_AVAIL bit will remain set as long as there still is unread host data in the buffer. Secondly, the availability of data will trigger the USB_SERIAL_JTAG_SERIAL_OUT_RECV_PKT_INT interrupt. When data is available, it can be read by firmware through repeatedly reading bytes from USB_SERIAL_JTAG_EP1_REG. The amount of bytes to read can be determined by checking the USB_SERIAL_JTAG_SERIAL_OUT_EP_DATA_AVAIL bit after reading each byte to see if there is more data to read. After all data is read, the USB device is automatically readied to receive a new data packet from the host.

When the firmware has data to send, it can put the data in the send buffer and trigger a flush to allow the host to receive the data in a USB packet. In order to do so, there needs to be space available in the send buffer. Firmware can check this by reading USB_REG_SERIAL_IN_EP_DATA_FREE. A 1 in this register field indicates there is still free room in the buffer, and firmware can fill the buffer by writing bytes to the USB_SERIAL_JTAG_EP1_REG register. Writing the buffer does not immediately trigger sending data to the host until the buffer is flushed. After the flush, the entire buffer will be ready to be received by the USB host at once. A flush can be triggered in two ways: 1) after the 64th byte is written to the buffer, the USB hardware will automatically flush the buffer to the host; or 2) firmware can trigger a flush by writing 1 to USB_REG_SERIAL_WR_DONE.

Regardless of how a flush is triggered, the send buffer will be unavailable for firmware to write into until it has

been fully read by the host. As soon as the send buffer has been fully read, the USB_SERIAL_JTAG_SERIAL_IN_EMPTY_INT interrupt will be triggered, indicating that the send buffer can receive another 64 bytes.

It is possible to handle some out-of-band serial requests in software, specifically, the host setting DTR and RTS and changing the line state. If the CDC-ACM interface receives a SET_LINE_CODING request, the peripheral can be configured to trigger a USB_SERIAL_JTAG_SET_LINE_CODE_INT interrupt, at which point the line coding can be read from the USB_SERIAL_JTAG_SET_LINE_CODE_WO_REG register. Similarly, SET_CONTROL_LINE_STATE requests will trigger USB_SERIAL_JTAG_RTS_CHG_INT and USB_SERIAL_JTAG_DTR_CHG_INT interrupts if they change the state of these lines. Software can then read the specific state through the USB_SERIAL_JTAG_RTS and USB_SERIAL_JTAG_DTR bits. Note that as described earlier, certain RTS/DTR sequences lead to hardware reset of ESP32-P4. Software can disable hardware recognition of these DTR/RTS sequences by setting the USB_SERIAL_JTAG_USB_UART_CHIP_RST_DIS bit, allowing software to interpret these signals freely.

Finally, the host can read the current line state using GET_LINE_CODING. This event sends back the data in the USB_SERIAL_JTAG_GET_LINE_CODE_WO_REG register and triggers a USB_SERIAL_JTAG_GET_LINE_CODE_INT interrupt.

51.3.3 USB-to-JTAG Interface: JTAG Command Processor

The USB-to-JTAG interface uses a vendor-specific class for its implementation. It consists of two endpoints, one to receive commands and another to send responses. Additionally, some less time-sensitive commands can be given as control requests.

Commands from the host to the JTAG interface are interpreted by the JTAG command processor. Internally, the JTAG command processor implements a full four-wire JTAG bus, consisting of the TCK, TMS and TDI output lines to the RISC-V CPU, as well as the TDO line signalling back from the CPU to the JTAG response capture unit. These signals adhere to the IEEE 1149.1 JTAG standards. Additionally, there is an SRST line to reset ESP32-P4.

Optionally, software can set USB_SERIAL_JTAG_USB_JTAG_BRIDGE_EN in order to redirect these signals to the GPIO matrix instead, where they can be routed to IO pads on ESP32-P4. This also allows external devices to be debugged via the USB Serial/JTAG peripheral.

The JTAG command processor parses each received nibble (4-bit value) as a command. As USB data is received in 8-bit bytes, this means each byte contains two commands. The USB command processor will execute high-nibble first and low-nibble second. The commands are used to control the TCK, TMS, TDI, and SRST lines of the internal JTAG bus, as well as to signal the JTAG response capture unit the state of the TDO line (which is driven by the CPU debugging logic) that needs to be captured.

In the internal JTAG bus, TCK, TMS, TDI, and TDO are connected directly to the JTAG debugging logic of the RISC-V CPU. SRST is connected to the reset logic of the digital circuitry in ESP32-P4 and a high level on this line will cause a digital system reset. Note that the USB Serial/JTAG controller itself is not affected by SRST.

A nibble can contain the following commands:

Table 51.3-3. Commands of a Nibble

bit	3	2	1	0
CMD_CLK	0	cap	tms	tdi
CMD_RST	1	0	0	srst
CMD_FLUSH	1	0	1	0
CMD_RSV	1	0	1	1
CMD_REP	1	1	R1	R0

- CMD_CLK will set the TDI and TMS as the indicated values and emit one clock pulse on TCK. If the CAP bit is 1, it will instruct the JTAG response capture unit to capture the state of the TDO line. This instruction forms the basis of JTAG communication.
- CMD_RST will set the state of the SRST line as the indicated value. This can be used to reset ESP32-P4.
- CMD_FLUSH will instruct the JTAG response capture unit to flush the buffer of all bits it collected so the host is able to read them. Note that in some cases, a JTAG transaction will end in an odd number of commands and as such an odd number of nibbles. In this case, it is allowed to repeat the CMD_FLUSH command to get an even number of nibbles fitting an integer number of bytes.
- CMD_RSV is reserved in the current implementation. This command will be ignored when received by ESP32-P4.
- CMD_REP repeats the last (non-CMD_REP) command for a certain number of times. The purpose is to compress command streams which repeat the CMD_CLK instruction for multiple times. A command such as CMD_CLK can be followed by multiple CMD_REP commands. The number of repetitions done by one CMD_REP can be expressed as $repetition\_count = (R1 \times 2 + R0) \times (4^{cmd\_rep\_count})$, where cmd_rep_count indicates the number of the CMD_REP instruction that went directly before it. Note that the CMD_REP command is only intended to repeat a CMD_CLK command. Specifically, using it on a CMD_FLUSH command may lead to an unresponsive USB device, and a USB reset will be required to recover it.

51.3.4 USB-to-JTAG Interface: CMD_REP Usage Example

Here is a list of commands as an illustration of the usage of CMD_REP. Note that each command is a nibble, and in this example, the bytewise command stream would be 0x0D 0x5E 0xCF.

1. 0x0 (CMD_CLK: cap=0, tdi=0, tms=0)
2. 0xD (CMD_REP: R1=0, R0=1)
3. 0x5 (CMD_CLK: cap=1, tdi=0, tms=1)
4. 0xE (CMD_REP: R1=1, R0=0)
5. 0xC (CMD_REP: R1=0, R0=0)
6. 0xF (CMD_REP: R1=1, R0=1)

The following shows what happens at every step:

1. TCK is clocked with the TDI and TMS lines set to 0. No data is captured.
2. TCK is clocked another $(0 \times 2 + 1) \times (4^0) = 1$ time with the same settings as step 1.

3. TCK is clocked with the TDI line set to 0 and TMS set to 1. Data on the TDO line is captured.
4. TCK is clocked another $(1 \times 2 + 0) \times (4^0) = 2$ times with the same settings as step 3.
5. Nothing happens: $(0 \times 2 + 0) \times (4^1) = 0$. Note that this increases cmd_rep_count in the next step.
6. TCK is clocked another $(1 \times 2 + 1) \times (4^2) = 48$ times with the same settings as step 3.

In other words, this example stream has the same net effect as that of executing command 1 twice, then repeating command 3 for 51 times.

51.3.5 USB-to-JTAG Interface: Response Capture Unit

The response capture unit reads the TDO line of the internal JTAG bus and captures its value when the command parser executes a CMD_CLK with cap=1. It puts this bit into an internal shift register, and writes a byte into the USB buffer when 8 bits have been collected. Of these 8 bits, the least significant one is the one that is read from TDO the earliest.

As soon as either 64 bytes (512 bits) have been collected or a CMD_FLUSH command is executed, the response capture unit will make the buffer available for the host to receive. Note that the interface to the USB logic is double-buffered. Therefore, as long as the USB throughput is sufficient, the response capture unit can always receive more data. That is to say, while one of the buffers is waiting to be sent to the host, the other can receive more data. When the host has received data from its buffer and the response capture unit flushes its buffer, the two buffers exchange position.

This also means that a command stream can cause at most 128 bytes of capture data generated (less if there are flush commands in the stream) without the host acting to receive the generated data. If more data is generated anyway, the command stream will pause and the device will not accept more commands until the generated capture data is read out.

Note that in general, the logic of the response capture unit tries not to send zero-byte responses. For instance, sending a series of CMD_FLUSH commands will not cause a series of 0-byte USB responses to be sent. However, in the current implementation, some zero-0 responses may be generated in extraordinary circumstances. It is recommended to ignore these responses.

51.3.6 USB-to-JTAG Interface: Control Transfer Requests

Aside from the command processor and the response capture unit, the USB-to-JTAG interface also understands some control requests, as documented in the table below:

Table 51.3-4. USB-to-JTAG Control Requests

bmRequestType	bRequest	wValue	wIndex	wLength	Data
01000000b	0 (VEND_JTAG_SETDIV)	[divider]	interface	0	None
01000000b	1 (VEND_JTAG_SETIO)	[iobits]	interface	0	None
11000000b	2 (VEND_JTAG_GETTDO)	0	interface	1	[iostate]
10000000b	6 (GET_DESCRIPTOR)	0x2000	0	256	[jtag cap desc]

- VEND_JTAG_SETDIV sets the divider used. This directly affects the duration of a TCK clock pulse. The TCK clock pulses are derived from a base clock of 48 MHz, which is divided down using an internal

divider. This control request allows the host to set this divider. Note that on startup, the divider is set to 2, which means the TCK clock rate will generally be 24 MHz.

- VEND_JTAG_SETIO can bypass the JTAG command processor to set the internal TDI, TDO, TMS, and SRST lines to given values. These values are encoded in the wValue field in the format of 11'b0, srst, trst, tck, tms, tdi.
- VEND_JTAG_GETTDO can bypass the JTAG response capture unit to read the internal TDO signal directly. This request returns one byte of data, of which the least significant bit represents the status of the TDO line.
- GET_DESCRIPTOR is a standard USB request. However, it can also be used with a vendor-specific wValue of 0x2000 to get the JTAG capabilities descriptor. This returns a certain amount of bytes representing the following fixed structure, which describes the capabilities of the USB-to-JTAG adapter (as shown in Table 51.3-5). This structure allows host software to automatically support future revisions of the hardware without the need for an update.

The JTAG capability descriptors of ESP32-P4 are as follows. Note that all 16-bit values are little-endian.

Table 51.3-5. JTAG Capability Descriptors

Byte	Value	Description
0	1	JTAG protocol capability structure version
1	10	Total length of JTAG protocol capabilities
2	1	Type of this struct: 1 for speed capability struct
3	8	Length of this speed capabilities struct
4 ~ 5	4800	JTAG base clock speed in 10 kHz increments. Note that the maximum TCK speed is half of this value
6 ~ 7	1	Minimum divider value settable by the VEND_JTAG_SETDIV request
8 ~ 9	255	Maximum divider value settable by the VEND_JTAG_SETDIV request

51.4 Recommended Operation

Little setup is needed for using the USB Serial/JTAG device. The USB-to-JTAG hardware itself does not need any setup aside from the standard USB initialization that the host operating system already does. Apart from that, the CDC-ACM emulation on the host side is also plug-and-play.

On the firmware side, very little initialization is needed either. The USB hardware is self-initialized and after boot-up, if a host is connected and listening on the CDC-ACM interface, data can be exchanged as described above without any specific setup except for the situation when the firmware optionally sets up an interrupt service handler.

One thing to note is that there may be situations where either the host is not attached or the CDC-ACM virtual port is not opened. In such cases, the packets that are flushed to the host will never be picked up and the send buffer will never be empty. It is important to detect these situations and implement timeout, as this is the only way to reliably detect whether the port on the host side is closed or not.

Another thing to note is that the USB device is dependent on the BBPLL for the 48 MHz USB PHY clock. If this PLL is disabled, the USB communication will cease to function.

One scenario where this happens is Deep-sleep. The USB Serial/JTAG controller (as well as the attached RISC-V CPU) will be entirely powered down in Deep-sleep mode. If a device needs to be debugged in this mode, it may be preferable to use an external JTAG debugger and a serial interface instead.

The CDC-ACM interface can also be used to reset the SoC and take it into or out of download mode. Generating the correct sequence of handshake signals can be a bit complicated, since most operating systems only allow setting or resetting DTR and RTS separately, but not in tandem. Additionally, some drivers (e.g., the standard CDC-ACM driver on Windows) do not set DTR until RTS is set and the user needs to explicitly set RTS in order to 'propagate' the DTR value. The recommended procedures are introduced below.

To reset the SoC into download mode:

Table 51.4-1. Reset SoC into Download Mode

Action	Internal state	Note
Clear DTR	RTS=?, DTR=0	Initialize to known values
Clear RTS	RTS=0, DTR=0	-
Set DTR	RTS=0, DTR=1	Set download mode flag
Clear RTS	RTS=0, DTR=1	Propagate DTR
Set RTS	RTS=1, DTR=1	-
Clear DTR	RTS=1, DTR=0	Reset SoC
Set RTS	RTS=1, DTR=0	Propagate DTR
Clear RTS	RTS=0, DTR=0	Clear download flag

To reset the SoC into booting from flash:

Table 51.4-2. Reset SoC into Booting from flash

Action	Internal state	Note
Clear DTR	RTS=?, DTR=0	-
Clear RTS	RTS=0, DTR=0	Clear download flag
Set RTS	RTS=1, DTR=0	Reset SoC
Clear RTS	RTS=0, DTR=0	Exit reset

51.5 Interrupts

ESP32-P4's USB Serial/JTAG Controller can generate the USB_SERIAL_JTAG_INTR interrupt signal that will be sent to the Interrupt Matrix. There are several internal interrupt sources from the USB Serial/JTAG Controller that can generate this interrupt signal.

- USB_SERIAL_JTAG_JTAG_IN_FLUSH_INT: triggered when flush cmd is received for the JTAG bulk out endpoint.
- USB_SERIAL_JTAG_SOF_INT: triggered when SOF frame is received.
- USB_SERIAL_JTAG_SERIAL_OUT_RECV_PKT_INT: triggered when Serial Port OUT Endpoint receives one packet.

- USB_SERIAL_JTAG_SERIAL_IN_EMPTY_INT: triggered when Serial Port IN Endpoint is empty.
- USB_SERIAL_JTAG_PID_ERR_INT: triggered when PID error is detected.
- USB_SERIAL_JTAG_CRC5_ERR_INT: triggered when CRC5 error is detected.
- USB_SERIAL_JTAG_CRC16_ERR_INT: triggered when CRC16 error is detected.
- USB_SERIAL_JTAG_STUFF_ERR_INT: triggered when a bit stuffing error is detected.
- USB_SERIAL_JTAG_IN_TOKEN_REC_IN_EP1_INT: triggered when IN token for IN endpoint 1 is received.
- USB_SERIAL_JTAG_USB_BUS_RESET_INT: triggered when USB bus reset is detected.
- USB_SERIAL_JTAG_OUT_EP1_ZERO_PAYLOAD_INT: triggered when OUT endpoint 1 receives packet with zero payload.
- USB_SERIAL_JTAG_OUT_EP2_ZERO_PAYLOAD_INT: triggered when OUT endpoint 2 receives packet with zero payload.
- USB_SERIAL_JTAG_RTS_CHG_INT: triggered when level of RTS from USB serial channel is changed.
- USB_SERIAL_JTAG_DTR_CHG_INT: triggered when level of DTR from USB serial channel is changed.
- USB_SERIAL_JTAG_GET_LINE_CODE_INT: triggered when level of GET LINE CODING request is received.
- USB_SERIAL_JTAG_SET_LINE_CODE_INT: triggered when level of SET LINE CODING request is received.

51.6 Register Summary

The addresses in this section are relative to USB Serial/JTAG controller base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

Name	Description	Address	Access
Configuration Registers			
USB_SERIAL_JTAG_EP1_REG	FIFO access for the CDC-ACM data IN and OUT endpoints	0x0000	R/W
USB_SERIAL_JTAG_EP1_CONF_REG	Configuration and control registers for the CDC-ACM FIFOs	0x0004	varies
USB_SERIAL_JTAG_CONF0_REG	PHY hardware configuration	0x0018	R/W
USB_SERIAL_JTAG_TEST_REG	Registers used for debugging the PHY	0x001C	varies
USB_SERIAL_JTAG_MISC_CONF_REG	Clock enable control	0x0044	R/W
USB_SERIAL_JTAG_MEM_CONF_REG	Memory power control	0x0048	R/W
USB_SERIAL_JTAG_CHIP_RST_REG	CDC-ACM chip reset control	0x004C	varies
USB_SERIAL_JTAG_GET_LINE_CODE_W0_REG	W0 of GET_LINE_CODING command	0x0058	R/W
USB_SERIAL_JTAG_GET_LINE_CODE_W1_REG	W1 of GET_LINE_CODING command	0x005C	R/W
USB_SERIAL_JTAG_CONFIG_UPDATE_REG	Configuration registers' value update	0x0060	WT
USB_SERIAL_JTAG_SER_AFIFO_CONFIG_REG	Serial AFIFO configure register	0x0064	varies
Interrupt Registers			
USB_SERIAL_JTAG_INT_RAW_REG	Interrupt raw status register	0x0008	R/WTC/SS
USB_SERIAL_JTAG_INT_ST_REG	Interrupt status register	0x000C	RO
USB_SERIAL_JTAG_INT_ENA_REG	Interrupt enable status register	0x0010	R/W
USB_SERIAL_JTAG_INT_CLR_REG	Interrupt clear status register	0x0014	WT
Status Registers			
USB_SERIAL_JTAG_JFIFO_ST_REG	JTAG FIFO status and control registers	0x0020	varies
USB_SERIAL_JTAG_FRAM_NUM_REG	Last received SOF frame index register	0x0024	RO
USB_SERIAL_JTAG_IN_EP0_ST_REG	Control IN endpoint status information	0x0028	RO
USB_SERIAL_JTAG_IN_EP1_ST_REG	CDC-ACM IN endpoint status information	0x002C	RO
USB_SERIAL_JTAG_IN_EP2_ST_REG	CDC-ACM interrupt IN endpoint status information	0x0030	RO
USB_SERIAL_JTAG_IN_EP3_ST_REG	JTAG IN endpoint status information	0x0034	RO
USB_SERIAL_JTAG_OUT_EP0_ST_REG	Control OUT endpoint status information	0x0038	RO
USB_SERIAL_JTAG_OUT_EP1_ST_REG	CDC-ACM OUT endpoint status information	0x003C	RO
USB_SERIAL_JTAG_OUT_EP2_ST_REG	JTAG OUT endpoint status information	0x0040	RO

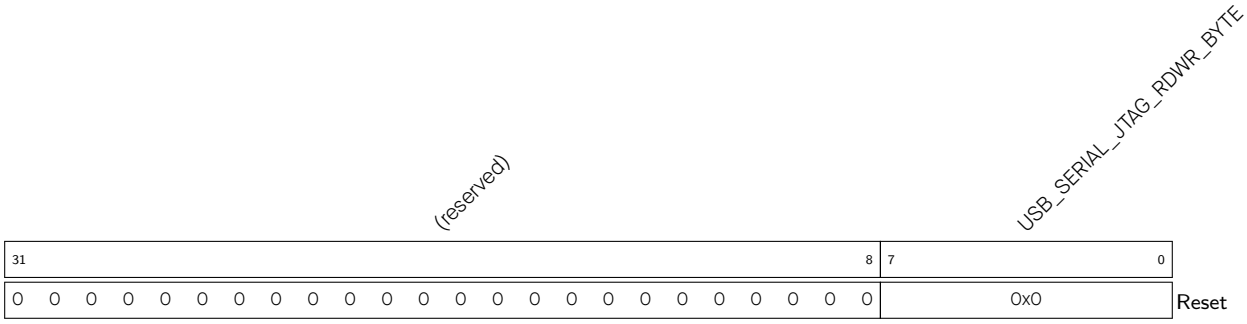
Name	Description	Address	Access
USB_SERIAL_JTAG_SET_LINE_CODE_W0_REG	W0 of SET_LINE_CODING command	0x0050	RO
USB_SERIAL_JTAG_SET_LINE_CODE_W1_REG	W1 of SET_LINE_CODING command	0x0054	RO
USB_SERIAL_JTAG_BUS_RESET_ST_REG	USB Bus reset status register	0x0068	RO
Version Registers			
USB_SERIAL_JTAG_DATE_REG	Date register	0x0088	R/W

51.7 Registers

The addresses in this section are relative to USB Serial/JTAG controller base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

For how to program reserved fields, please refer to Section Programming Reserved Register Field.

Register 51.1. USB_SERIAL_JTAG_EP1_REG (0x0000)



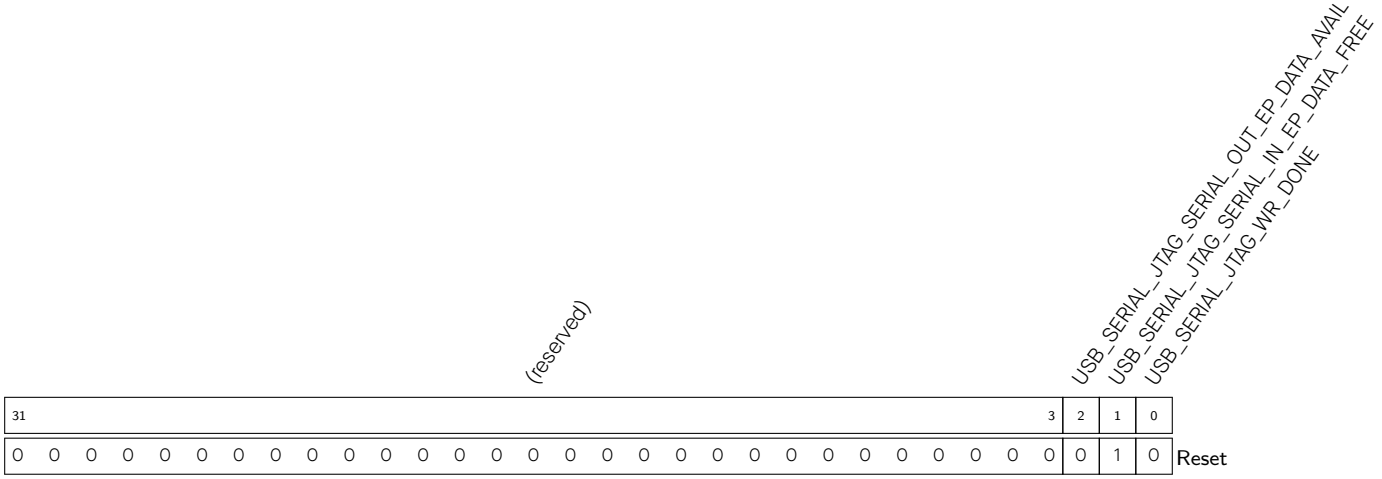
USB_SERIAL_JTAG_RDWR_BYTE A write to this register pushes the written data into the CDC TX FIFO; a read from this register pops a byte from the CDC RX FIFO and returns it.

When USB_SERIAL_JTAG_SERIAL_IN_EMPTY_INT is set, users can write data (up to 64 bytes) into CDC-ACM TX FIFO through this register.

When USB_SERIAL_JTAG_SERIAL_OUT_RECV_PKT_INT is set, users can check how many data is received through USB_SERIAL_JTAG_OUT_EP1_WR_ADDR, then read data from CDC-ACM RX FIFO through this register.

(R/W)

Register 51.2. USB_SERIAL_JTAG_EP1_CONF_REG (0x0004)



USB_SERIAL_JTAG_WR_DONE Configures whether to represent writing byte data to CDC-ACM TX FIFO is done.

0: No effect

1: Represents writing byte data to CDC-ACM TX FIFO is done

This bit then stays 0 until data in CDC-ACM TX FIFO is read by the USB Host.

(WT)

USB_SERIAL_JTAG_SERIAL_IN_EP_DATA_FREE Represents whether CDC-ACM TX FIFO has space available.

0: CDC-ACM TX FIFO is full and no data should be written into it

1: CDC-ACM TX FIFO is not full and data can be written into it

After writing USB_SERIAL_JTAG_WR_DONE, this bit will be 0 until data in CDC-ACM TX FIFO is read by USB Host.

(RO)

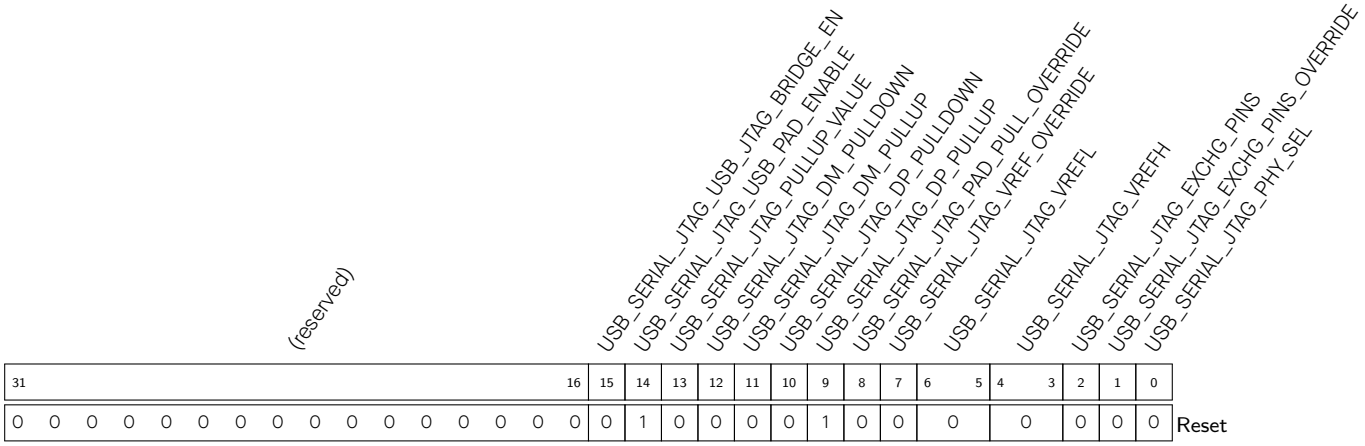
USB_SERIAL_JTAG_SERIAL_OUT_EP_DATA_AVAIL Represents whether there is data in CDC-ACM RX FIFO.

0: There is no data in CDC-ACM RX FIFO

1: There is data in CDC-ACM RX FIFO

(RO)

Register 51.3. USB_SERIAL_JTAG_CONF0_REG (0x0018)



USB_SERIAL_JTAG_PHY_SEL Configures whether to select internal or external PHY.

0: Internal PHY

1: External PHY

(R/W)

USB_SERIAL_JTAG_EXCHG_PINS_OVERRIDE Configures whether to enable software control USB D+ D- exchange.

0: Disable

1: Enable

(R/W)

USB_SERIAL_JTAG_EXCHG_PINS Configures whether to enable USB D+ D- exchange.

0: Disable

1: Enable

(R/W)

USB_SERIAL_JTAG_VREFH Configures single-end input high threshold.

0: 1.76 V

1: 1.84 V

2: 1.92 V

3: 2.00 V

(R/W)

USB_SERIAL_JTAG_VREFL Configures single-end input low threshold.

0: 0.80 V

1: 0.88 V

2: 0.96 V

3: 1.04 V

(R/W)

USB_SERIAL_JTAG_VREF_OVERRIDE Configures whether to enable software control input threshold.

0: Disable

1: Enable

(R/W)

Continued on the next page...

Register 51.3. USB_SERIAL_JTAG_CONF0_REG (0x0018)

Continued from the previous page...

USB_SERIAL_JTAG_PAD_PULL_OVERRIDE Configures whether to enable software to control USB D+ D- pullup and pulldown.
0: Disable
1: Enable
(R/W)

USB_SERIAL_JTAG_DP_PULLUP Configures whether to enable USB D+ pull up when USB_SERIAL_JTAG_PAD_PULL_OVERRIDE is 1.
0: Disable
1: Enable
(R/W)

USB_SERIAL_JTAG_DP_PULLDOWN Configures whether to enable USB D+ pull down when USB_SERIAL_JTAG_PAD_PULL_OVERRIDE is 1.
0: Disable
1: Enable
(R/W)

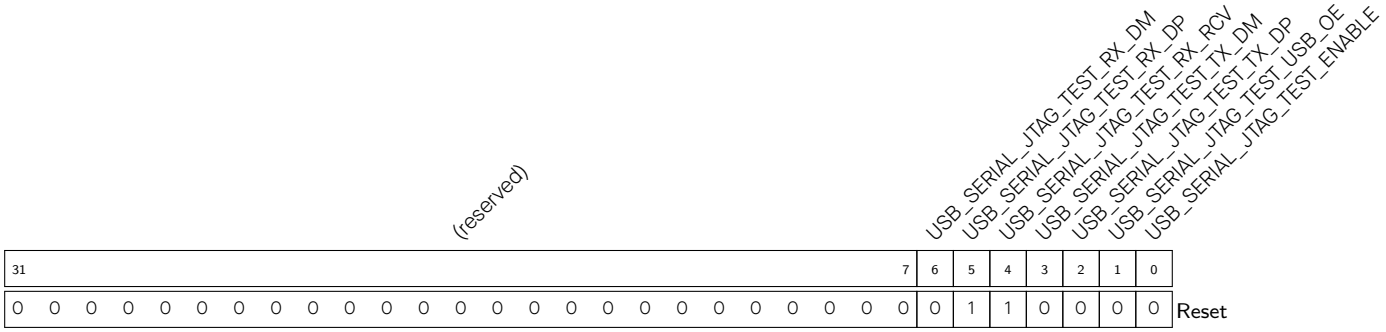
USB_SERIAL_JTAG_DM_PULLDOWN Configures whether to enable USB D- pull down when USB_SERIAL_JTAG_PAD_PULL_OVERRIDE is 1.
0: Disable
1: Enable
(R/W)

USB_SERIAL_JTAG_PULLUP_VALUE Configures the pull up value when USB_SERIAL_JTAG_PAD_PULL_OVERRIDE is 1.
0: 2.2 K
1: 1.1 K
(R/W)

USB_SERIAL_JTAG_USB_PAD_ENABLE Configures whether to enable USB pad function.
0: Disable
1: Enable
(R/W)

USB_SERIAL_JTAG_USB_JTAG_BRIDGE_EN Configures whether to disconnect USB_JTAG and internal JTAG.
0: USB_JTAG is connected to the internal JTAG port of CPU
1: USB_JTAG and the internal JTAG are disconnected, MTMS, MTDI, and MTCK are output through GPIO Matrix, and MTDO is input through GPIO Matrix
(R/W)

Register 51.4. USB_SERIAL_JTAG_TEST_REG (0x001C)



USB_SERIAL_JTAG_TEST_ENABLE Configures whether to enable the test mode of the USB pad.

0: Resume normal operation

1: Enable the test mode of the USB pad

Enabling the test mode of the USB pad allows the USB pad to be controlled/read using the other bits in this register.

(R/W)

USB_SERIAL_JTAG_TEST_USB_OE Configures whether to enable USB pad output.

0: Set D+ and D- to high impedance

1: Output the values set in USB_SERIAL_JTAG_TEST_TX_DP and USB_SERIAL_JTAG_TEST_TX_DM on the D+ and D- pins

(R/W)

USB_SERIAL_JTAG_TEST_TX_DP Configures value of USB D+ in test mode when USB_SERIAL_JTAG_TEST_USB_OE is 1. (R/W)

USB_SERIAL_JTAG_TEST_TX_DM Configures value of USB D- in test mode when USB_SERIAL_JTAG_TEST_USB_OE is 1. (R/W)

USB_SERIAL_JTAG_TEST_RX_RCV Represents the current logical level of the voltage difference between USB D- and USB D+ pads in test mode.

0: USB D- voltage is higher than USB D+

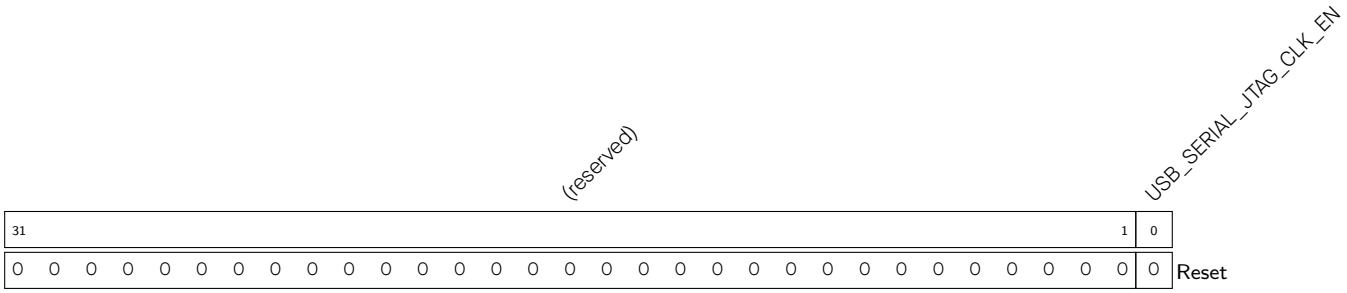
1: USB D+ voltage is higher than USB D-

(RO)

USB_SERIAL_JTAG_TEST_RX_DP Represents the logical level of the USB D+ pad in test mode. (RO)

USB_SERIAL_JTAG_TEST_RX_DM Represents the logical level of the USB D- pad in test mode. (RO)

Register 51.5. USB_SERIAL_JTAG_MISC_CONF_REG (0x0044)



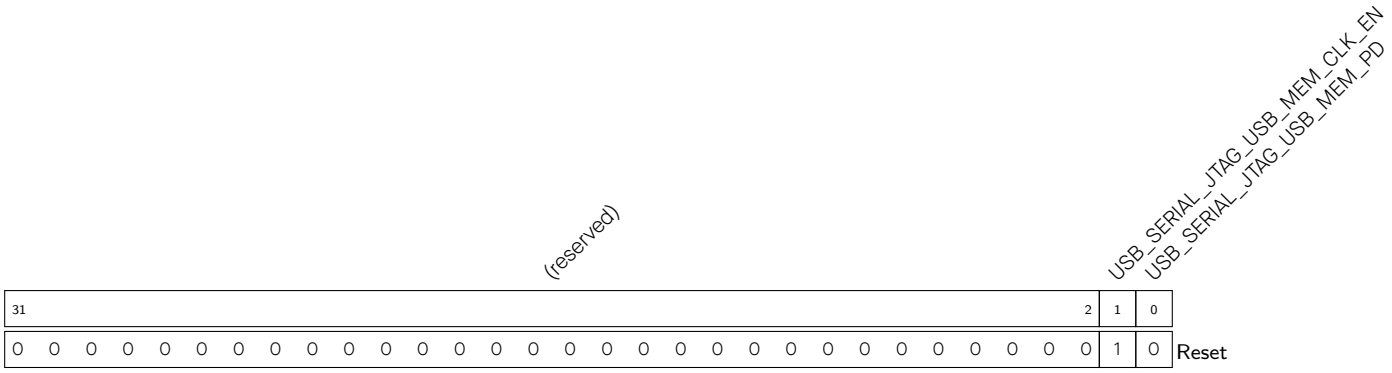
USB_SERIAL_JTAG_CLK_EN Configures whether to force clock on for register.

0: Support clock only when application writes registers

1: Force clock on for register

(R/W)

Register 51.6. USB_SERIAL_JTAG_MEM_CONF_REG (0x0048)



USB_SERIAL_JTAG_USB_MEM_PD Configures whether to power down USB memory.

0: No effect

1: Power down

(R/W)

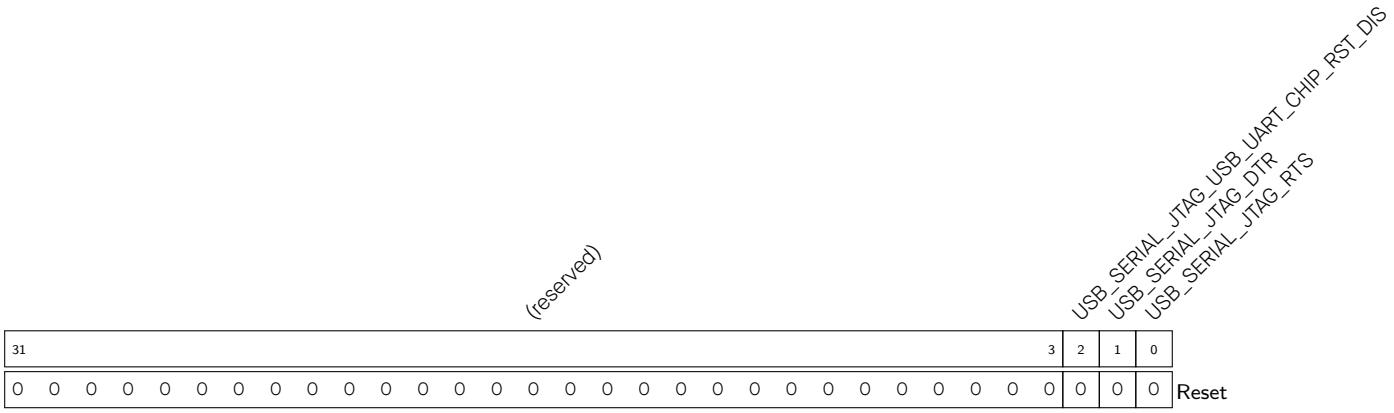
USB_SERIAL_JTAG_USB_MEM_CLK_EN Configures whether to force clock on for USB memory.

0: No effect

1: Force

(R/W)

Register 51.7. USB_SERIAL_JTAG_CHIP_RST_REG (0x004C)



USB_SERIAL_JTAG_RTS Represents the state of RTS signal as set by the most recent SET_LINE_CODING command. (RO)

USB_SERIAL_JTAG_DTR Represents the state of DTR signal as set by the most recent SET_LINE_CODING command. (RO)

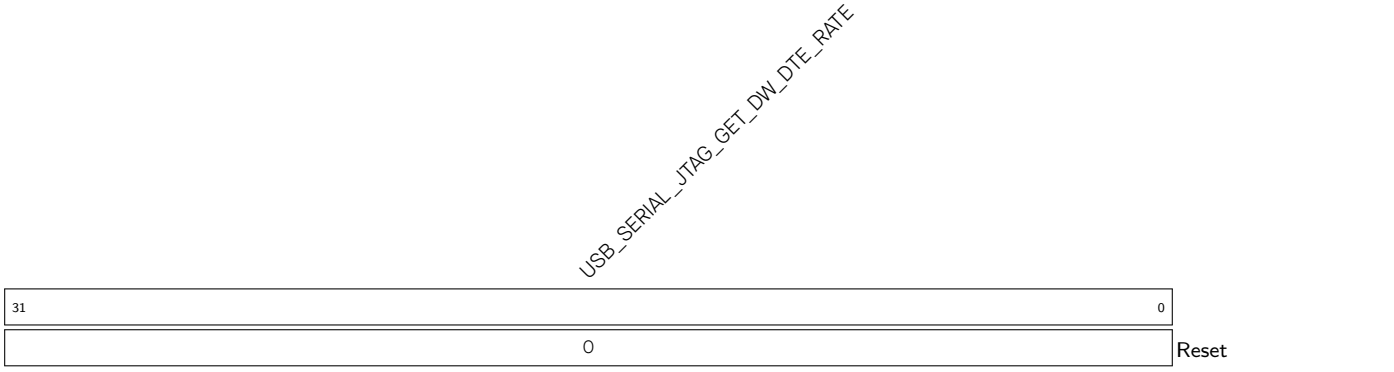
USB_SERIAL_JTAG_USB_UART_CHIP_RST_DIS Configures whether to disable chip reset from USB serial channel.

0: No effect

1: Disable

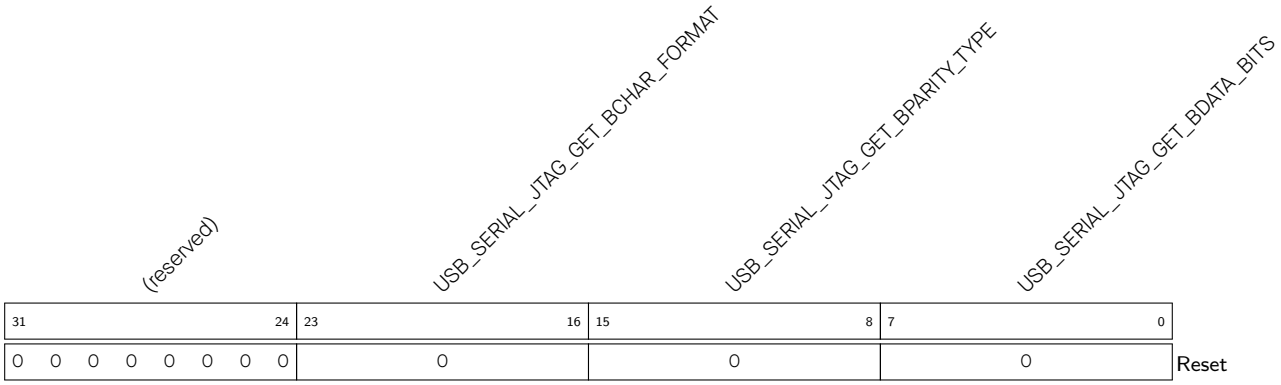
(R/W)

Register 51.8. USB_SERIAL_JTAG_GET_LINE_CODE_W0_REG (0x0058)



USB_SERIAL_JTAG_GET_DW_DTE_RATE Configures the value of dwDTERate set by software, which is requested by GET_LINE_CODING command. (R/W)

Register 51.9. USB_SERIAL_JTAG_GET_LINE_CODE_W1_REG (0x005C)

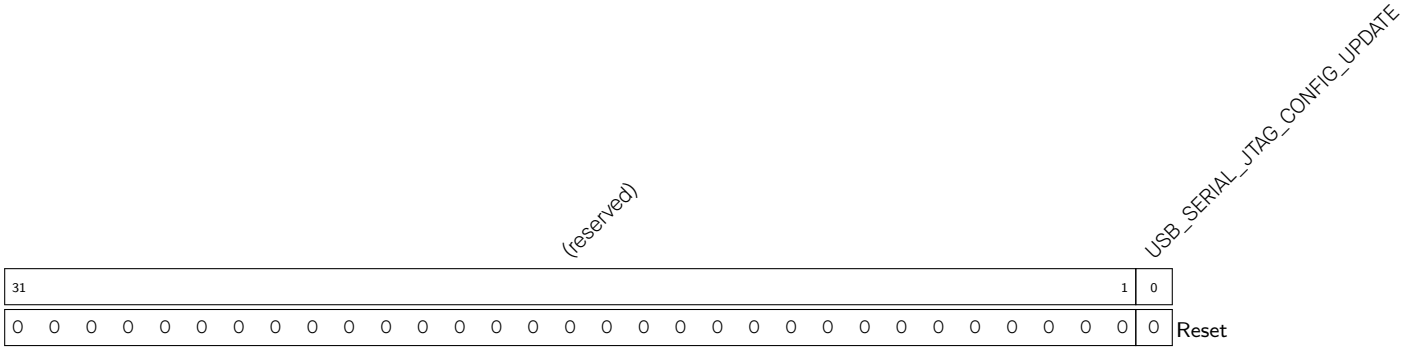


USB_SERIAL_JTAG_GET_BDATA_BITS Configures the value of bDataBits set by software, which is requested by GET_LINE_CODING command. (R/W)

USB_SERIAL_JTAG_GET_BPARITY_TYPE Configures the value of bParityType set by software, which is requested by GET_LINE_CODING command. (R/W)

USB_SERIAL_JTAG_GET_BCHAR_FORMAT Configures the value of bCharFormat set by software, which is requested by GET_LINE_CODING command. (R/W)

Register 51.10. USB_SERIAL_JTAG_CONFIG_UPDATE_REG (0x0060)



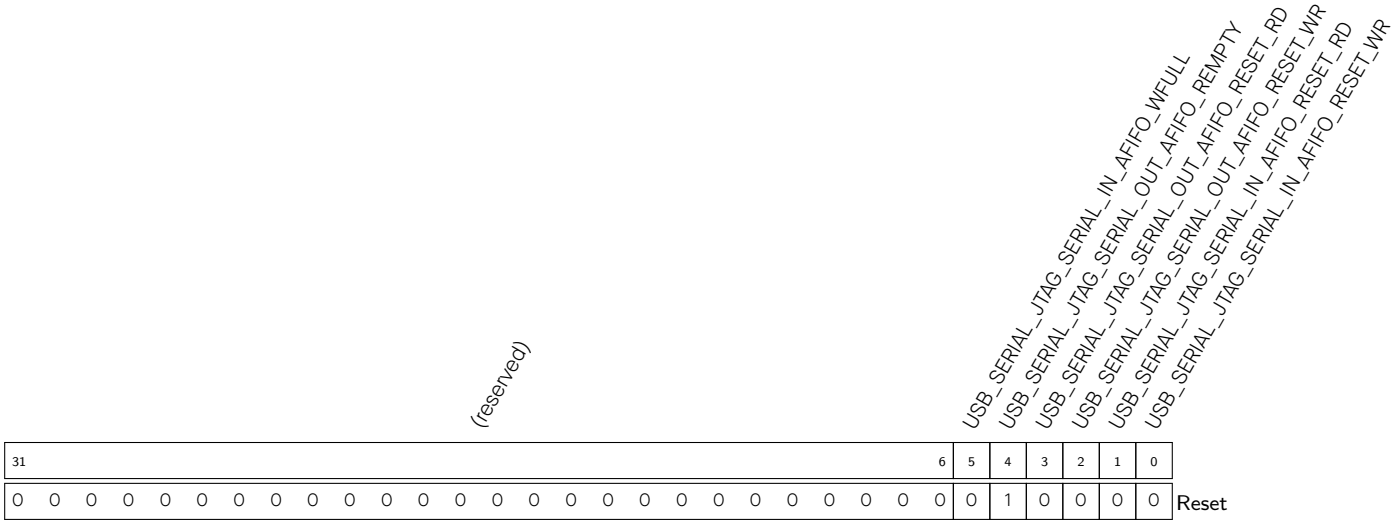
USB_SERIAL_JTAG_CONFIG_UPDATE Configures whether to update the value of configuration registers from APB clock domain to 48 MHz clock domain.

0: No effect

1: Update

(WT)

Register 51.11. USB_SERIAL_JTAG_SER_AFIFO_CONFIG_REG (0x0064)



USB_SERIAL_JTAG_SERIAL_IN_AFIFO_RESET_WR Configures whether to reset CDC_ACM IN async FIFO write clock domain.

0: No effect

1: Reset

(R/W)

USB_SERIAL_JTAG_SERIAL_IN_AFIFO_RESET_RD Configures whether to reset CDC_ACM IN async FIFO read clock domain.

0: No effect

1: Reset

(R/W)

USB_SERIAL_JTAG_SERIAL_OUT_AFIFO_RESET_WR Configures whether to reset CDC_ACM OUT async FIFO write clock domain.

0: No effect

1: Reset

(R/W)

USB_SERIAL_JTAG_SERIAL_OUT_AFIFO_RESET_RD Configures whether to reset CDC_ACM OUT async FIFO read clock domain.

0: No effect

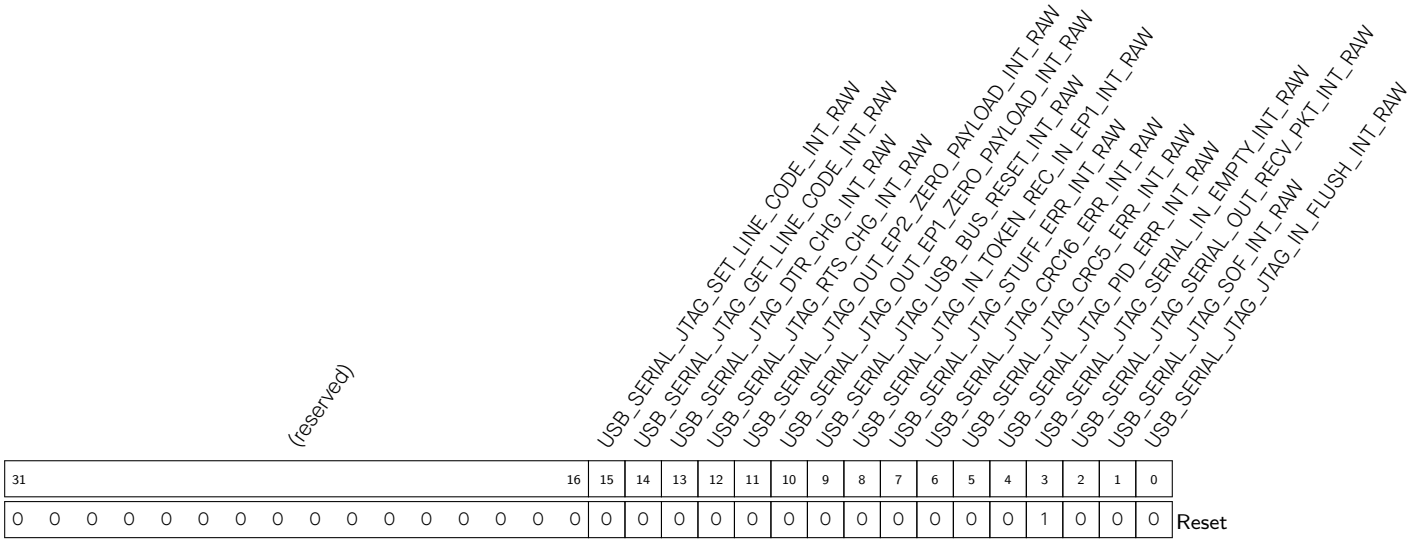
1: Reset

(R/W)

USB_SERIAL_JTAG_SERIAL_OUT_AFIFO_REMPTY Represents CDC_ACM OUT async FIFO empty signal in read clock domain. (RO)

USB_SERIAL_JTAG_SERIAL_IN_AFIFO_WFULL Represents CDC_ACM IN async FIFO full signal in write clock domain. (RO)

Register 51.12. USB_SERIAL_JTAG_INT_RAW_REG (0x0008)



USB_SERIAL_JTAG_JTAG_IN_FLUSH_INT_RAW The raw interrupt status of USB_SERIAL_JTAG_JTAG_IN_FLUSH_INT. (R/WTC/SS)

USB_SERIAL_JTAG_SOF_INT_RAW The raw interrupt status of USB_SERIAL_JTAG_SOF_INT. (R/WTC/SS)

USB_SERIAL_JTAG_SERIAL_OUT_RECV_PKT_INT_RAW The raw interrupt status of USB_SERIAL_JTAG_SERIAL_OUT_RECV_PKT_INT. (R/WTC/SS)

USB_SERIAL_JTAG_SERIAL_IN_EMPTY_INT_RAW The raw interrupt status of USB_SERIAL_JTAG_SERIAL_IN_EMPTY_INT. (R/WTC/SS)

USB_SERIAL_JTAG_PID_ERR_INT_RAW The raw interrupt status of USB_SERIAL_JTAG_PID_ERR_INT. (R/WTC/SS)

USB_SERIAL_JTAG_CRC5_ERR_INT_RAW The raw interrupt status of USB_SERIAL_JTAG_CRC5_ERR_INT. (R/WTC/SS)

USB_SERIAL_JTAG_CRC16_ERR_INT_RAW The raw interrupt status of USB_SERIAL_JTAG_CRC16_ERR_INT. (R/WTC/SS)

USB_SERIAL_JTAG_STUFF_ERR_INT_RAW The raw interrupt status of USB_SERIAL_JTAG_STUFF_ERR_INT. (R/WTC/SS)

USB_SERIAL_JTAG_IN_TOKEN_REC_IN_EP1_INT_RAW The raw interrupt status of USB_SERIAL_JTAG_IN_TOKEN_REC_IN_EP1_INT. (R/WTC/SS)

USB_SERIAL_JTAG_USB_BUS_RESET_INT_RAW The raw interrupt status of USB_SERIAL_JTAG_USB_BUS_RESET_INT. (R/WTC/SS)

USB_SERIAL_JTAG_OUT_EP1_ZERO_PAYLOAD_INT_RAW The raw interrupt status of USB_SERIAL_JTAG_OUT_EP1_ZERO_PAYLOAD_INT. (R/WTC/SS)

Continued on the next page...

Register 51.12. USB_SERIAL_JTAG_INT_RAW_REG (0x0008)

Continued from the previous page...

USB_SERIAL_JTAG_OUT_EP2_ZERO_PAYLOAD_INT_RAW The raw interrupt status of USB_SERIAL_JTAG_OUT_EP2_ZERO_PAYLOAD_INT. (R/WTC/SS)

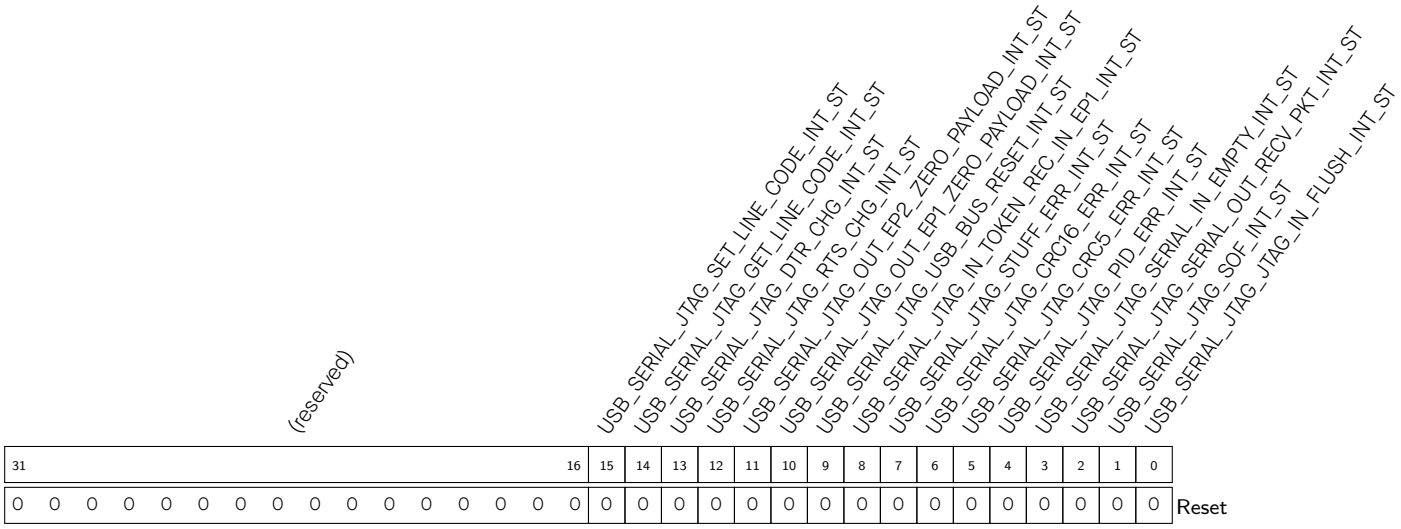
USB_SERIAL_JTAG_RTS_CHG_INT_RAW The raw interrupt status of USB_SERIAL_JTAG_RTS_CHG_INT. (R/WTC/SS)

USB_SERIAL_JTAG_DTR_CHG_INT_RAW The raw interrupt status of USB_SERIAL_JTAG_DTR_CHG_INT. (R/WTC/SS)

USB_SERIAL_JTAG_GET_LINE_CODE_INT_RAW The raw interrupt status of USB_SERIAL_JTAG_GET_LINE_CODE_INT. (R/WTC/SS)

USB_SERIAL_JTAG_SET_LINE_CODE_INT_RAW The raw interrupt status of USB_SERIAL_JTAG_SET_LINE_CODE_INT. (R/WTC/SS)

Register 51.13. USB_SERIAL_JTAG_INT_ST_REG (0x000C)



USB_SERIAL_JTAG_JTAG_IN_FLUSH_INT_ST The masked interrupt status of USB_SERIAL_JTAG_JTAG_IN_FLUSH_INT. (RO)

USB_SERIAL_JTAG_SOF_INT_ST The masked interrupt status of USB_SERIAL_JTAG_SOF_INT. (RO)

USB_SERIAL_JTAG_SERIAL_OUT_RECV_PKT_INT_ST The masked interrupt status of the USB_SERIAL_JTAG_SERIAL_OUT_RECV_PKT_INT. (RO)

USB_SERIAL_JTAG_SERIAL_IN_EMPTY_INT_ST The masked interrupt status of USB_SERIAL_JTAG_SERIAL_IN_EMPTY_INT. (RO)

USB_SERIAL_JTAG_PID_ERR_INT_ST The masked interrupt status of USB_SERIAL_JTAG_PID_ERR_INT. (RO)

USB_SERIAL_JTAG_CRC5_ERR_INT_ST The masked interrupt status of USB_SERIAL_JTAG_CRC5_ERR_INT. (RO)

USB_SERIAL_JTAG_CRC16_ERR_INT_ST The masked interrupt status of USB_SERIAL_JTAG_CRC16_ERR_INT. (RO)

USB_SERIAL_JTAG_STUFF_ERR_INT_ST The masked interrupt status of USB_SERIAL_JTAG_STUFF_ERR_INT. (RO)

USB_SERIAL_JTAG_IN_TOKEN_REC_IN_EP1_INT_ST The masked interrupt status of USB_SERIAL_JTAG_IN_TOKEN_REC_IN_EP1_INT. (RO)

USB_SERIAL_JTAG_USB_BUS_RESET_INT_ST The masked interrupt status of USB_SERIAL_JTAG_USB_BUS_RESET_INT. (RO)

USB_SERIAL_JTAG_OUT_EP1_ZERO_PAYLOAD_INT_ST The masked interrupt status of USB_SERIAL_JTAG_OUT_EP1_ZERO_PAYLOAD_INT (RO)

USB_SERIAL_JTAG_OUT_EP2_ZERO_PAYLOAD_INT_ST The masked interrupt status of USB_SERIAL_JTAG_OUT_EP2_ZERO_PAYLOAD_INT. (RO)

Continued on the next page...

Register 51.13. USB_SERIAL_JTAG_INT_ST_REG (0x000C)

Continued from the previous page...

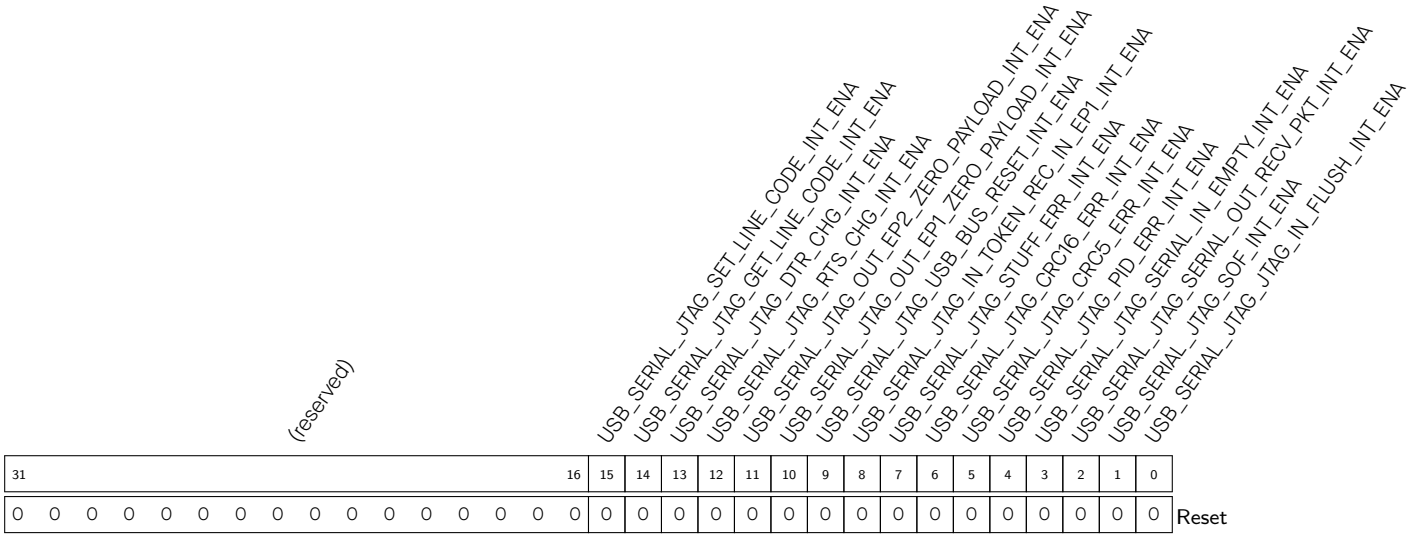
USB_SERIAL_JTAG_RTS_CHG_INT_ST The masked interrupt status of USB_SERIAL_JTAG_RTS_CHG_INT. (RO)

USB_SERIAL_JTAG_DTR_CHG_INT_ST The masked interrupt status of USB_SERIAL_JTAG_DTR_CHG_INT. (RO)

USB_SERIAL_JTAG_GET_LINE_CODE_INT_ST The masked interrupt status of USB_SERIAL_JTAG_GET_LINE_CODE_INT. (RO)

USB_SERIAL_JTAG_SET_LINE_CODE_INT_ST The masked interrupt status of USB_SERIAL_JTAG_SET_LINE_CODE_INT. (RO)

Register 51.14. USB_SERIAL_JTAG_INT_ENA_REG (0x0010)



USB_SERIAL_JTAG_JTAG_IN_FLUSH_INT_ENA Write 1 to enable USB_SERIAL_JTAG_JTAG_IN_FLUSH_INT. (R/W)

USB_SERIAL_JTAG_SOF_INT_ENA Write 1 to enable USB_SERIAL_JTAG_SOF_INT. (R/W)

USB_SERIAL_JTAG_SERIAL_OUT_RECV_PKT_INT_ENA Write 1 to enable USB_SERIAL_JTAG_SERIAL_OUT_RECV_PKT_INT. (R/W)

USB_SERIAL_JTAG_SERIAL_IN_EMPTY_INT_ENA Write 1 to enable USB_SERIAL_JTAG_SERIAL_IN_EMPTY_INT. (R/W)

USB_SERIAL_JTAG_PID_ERR_INT_ENA Write 1 to enable USB_SERIAL_JTAG_PID_ERR_INT. (R/W)

USB_SERIAL_JTAG_CRC5_ERR_INT_ENA Write 1 to enable USB_SERIAL_JTAG_CRC5_ERR_INT. (R/W)

USB_SERIAL_JTAG_CRC16_ERR_INT_ENA Write 1 to enable USB_SERIAL_JTAG_CRC16_ERR_INT. (R/W)

USB_SERIAL_JTAG_STUFF_ERR_INT_ENA Write 1 to enable USB_SERIAL_JTAG_STUFF_ERR_INT. (R/W)

USB_SERIAL_JTAG_IN_TOKEN_REC_IN_EP1_INT_ENA Write 1 to enable USB_SERIAL_JTAG_IN_TOKEN_REC_IN_EP1_INT. (R/W)

USB_SERIAL_JTAG_USB_BUS_RESET_INT_ENA Write 1 to enable USB_SERIAL_JTAG_USB_BUS_RESET_INT. (R/W)

USB_SERIAL_JTAG_OUT_EP1_ZERO_PAYLOAD_INT_ENA Write 1 to enable USB_SERIAL_JTAG_OUT_EP1_ZERO_PAYLOAD_INT. (R/W)

USB_SERIAL_JTAG_OUT_EP2_ZERO_PAYLOAD_INT_ENA Write 1 to enable USB_SERIAL_JTAG_OUT_EP2_ZERO_PAYLOAD_INT. (R/W)

Continued on the next page...

Register 51.14. USB_SERIAL_JTAG_INT_ENA_REG (0x0010)

Continued from the previous page...

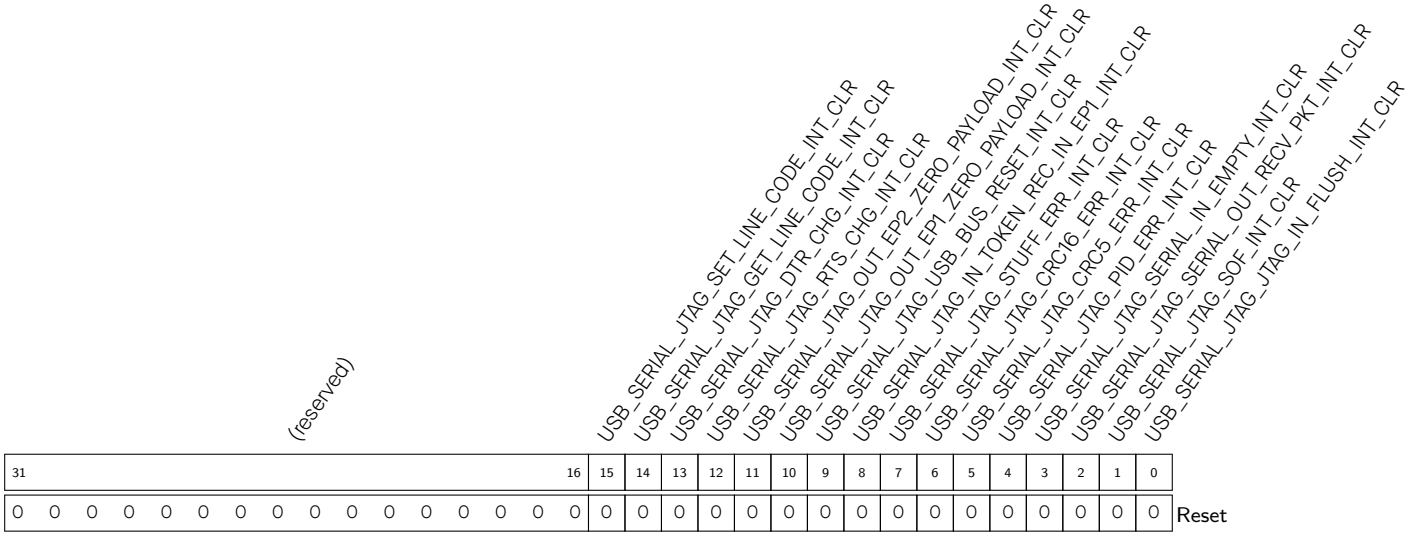
USB_SERIAL_JTAG_RTS_CHG_INT_ENA Write 1 to enable USB_SERIAL_JTAG_RTS_CHG_INT. (R/W)

USB_SERIAL_JTAG_DTR_CHG_INT_ENA Write 1 to enable USB_SERIAL_JTAG_DTR_CHG_INT. (R/W)

USB_SERIAL_JTAG_GET_LINE_CODE_INT_ENA Write 1 to enable USB_SERIAL_JTAG_GET_LINE_CODE_INT. (R/W)

USB_SERIAL_JTAG_SET_LINE_CODE_INT_ENA Write 1 to enable USB_SERIAL_JTAG_SET_LINE_CODE_INT. (R/W)

Register 51.15. USB_SERIAL_JTAG_INT_CLR_REG (0x0014)



USB_SERIAL_JTAG_JTAG_IN_FLUSH_INT_CLR Write 1 to clear USB_SERIAL_JTAG_JTAG_IN_FLUSH_INT. (WT)

USB_SERIAL_JTAG_SOF_INT_CLR Write 1 to clear USB_SERIAL_JTAG_SOF_INT. (WT)

USB_SERIAL_JTAG_SERIAL_OUT_RECV_PKT_INT_CLR Write 1 to clear USB_SERIAL_JTAG_SERIAL_OUT_RECV_PKT_INT. (WT)

USB_SERIAL_JTAG_SERIAL_IN_EMPTY_INT_CLR Write 1 to clear USB_SERIAL_JTAG_SERIAL_IN_EMPTY_INT. (WT)

USB_SERIAL_JTAG_PID_ERR_INT_CLR Write 1 to clear USB_SERIAL_JTAG_PID_ERR_INT. (WT)

USB_SERIAL_JTAG_CRC5_ERR_INT_CLR Write 1 to clear USB_SERIAL_JTAG_CRC5_ERR_INT. (WT)

USB_SERIAL_JTAG_CRC16_ERR_INT_CLR Write 1 to clear USB_SERIAL_JTAG_CRC16_ERR_INT. (WT)

USB_SERIAL_JTAG_STUFF_ERR_INT_CLR Write 1 to clear USB_SERIAL_JTAG_STUFF_ERR_INT. (WT)

USB_SERIAL_JTAG_IN_TOKEN_REC_IN_EP1_INT_CLR Write 1 to clear USB_SERIAL_JTAG_IN_TOKEN_REC_IN_EP1_INT. (WT)

USB_SERIAL_JTAG_USB_BUS_RESET_INT_CLR Write 1 to clear USB_SERIAL_JTAG_USB_BUS_RESET_INT. (WT)

USB_SERIAL_JTAG_OUT_EP1_ZERO_PAYLOAD_INT_CLR Write 1 to clear USB_SERIAL_JTAG_OUT_EP1_ZERO_PAYLOAD_INT. (WT)

USB_SERIAL_JTAG_OUT_EP2_ZERO_PAYLOAD_INT_CLR Write 1 to clear USB_SERIAL_JTAG_OUT_EP2_ZERO_PAYLOAD_INT. (WT)

Continued on the next page...

Register 51.15. USB_SERIAL_JTAG_INT_CLR_REG (0x0014)

Continued from the previous page...

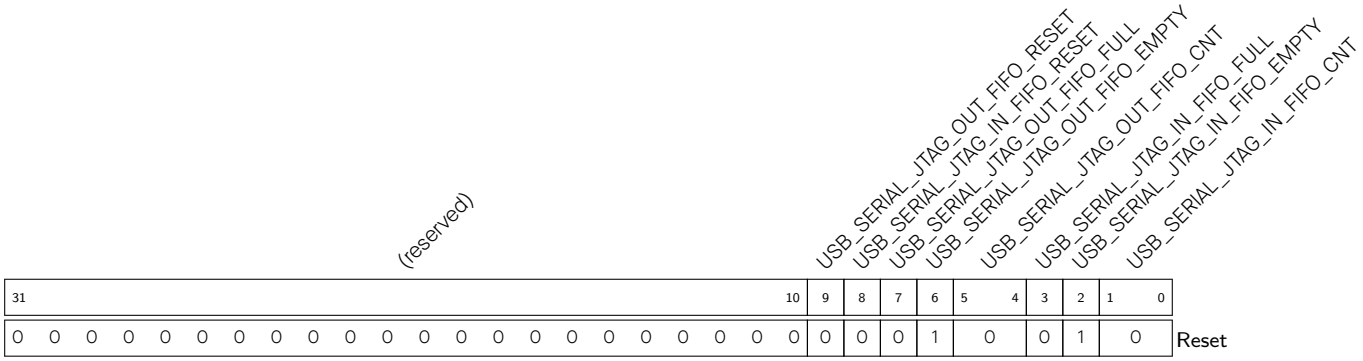
USB_SERIAL_JTAG_RTS_CHG_INT_CLR Write 1 to clear USB_SERIAL_JTAG_RTS_CHG_INT. (WT)

USB_SERIAL_JTAG_DTR_CHG_INT_CLR Write 1 to clear USB_SERIAL_JTAG_DTR_CHG_INT. (WT)

USB_SERIAL_JTAG_GET_LINE_CODE_INT_CLR Write 1 to clear USB_SERIAL_JTAG_GET_LINE_CODE_INT. (WT)

USB_SERIAL_JTAG_SET_LINE_CODE_INT_CLR Write 1 to clear USB_SERIAL_JTAG_SET_LINE_CODE_INT. (WT)

Register 51.16. USB_SERIAL_JTAG_JFIFO_ST_REG (0x0020)



USB_SERIAL_JTAG_IN_FIFO_CNT Represents JTAG IN FIFO counter. (RO)

USB_SERIAL_JTAG_IN_FIFO_EMPTY Represents whether JTAG IN FIFO is empty.

0: Not empty

1: Empty

(RO)

USB_SERIAL_JTAG_IN_FIFO_FULL Represents whether JTAG IN FIFO is full.

0: Not full

1: Full

(RO)

USB_SERIAL_JTAG_OUT_FIFO_CNT Represents JTAG OUT FIFO counter. (RO)

USB_SERIAL_JTAG_OUT_FIFO_EMPTY Represents whether JTAG OUT FIFO is empty.

0: Not empty

1: Empty

(RO)

USB_SERIAL_JTAG_OUT_FIFO_FULL Represents whether JTAG OUT FIFO is full.

0: Not full

1: Full

(RO)

USB_SERIAL_JTAG_IN_FIFO_RESET Configures whether to reset JTAG IN FIFO.

0: No effect

1: Reset

(R/W)

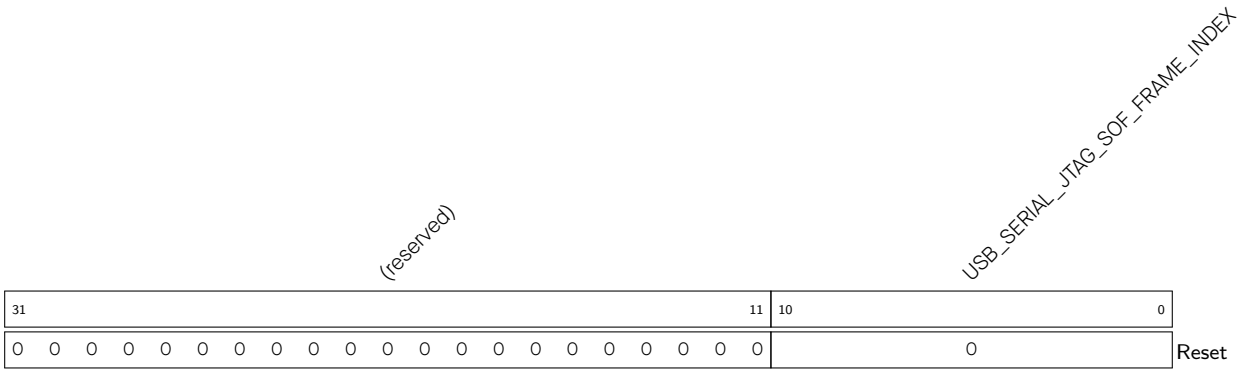
USB_SERIAL_JTAG_OUT_FIFO_RESET Configures whether to reset JTAG OUT FIFO.

0: No effect

1: Reset

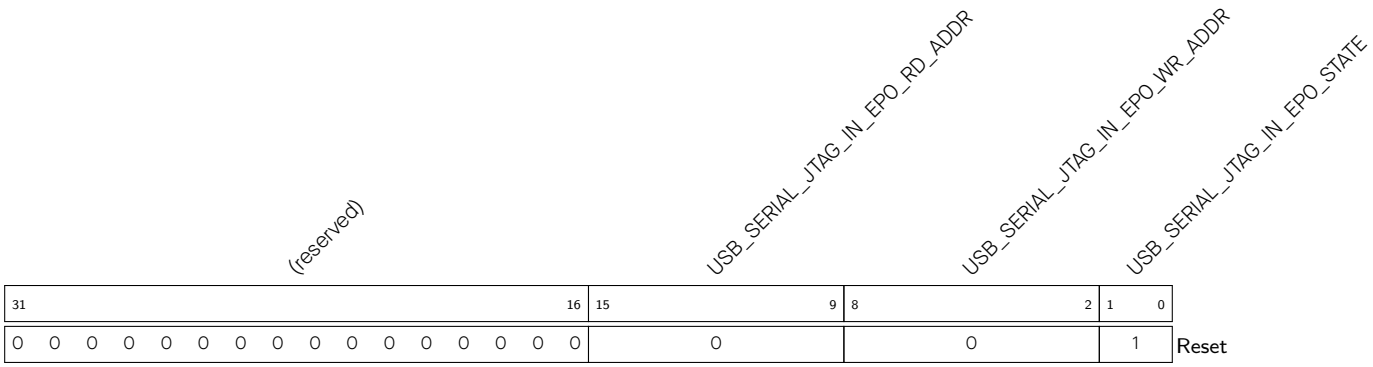
(R/W)

Register 51.17. USB_SERIAL_JTAG_FRAM_NUM_REG (0x0024)



USB_SERIAL_JTAG_SOF_FRAME_INDEX Represents frame index of received SOF frame. (RO)

Register 51.18. USB_SERIAL_JTAG_IN_EP0_ST_REG (0x0028)

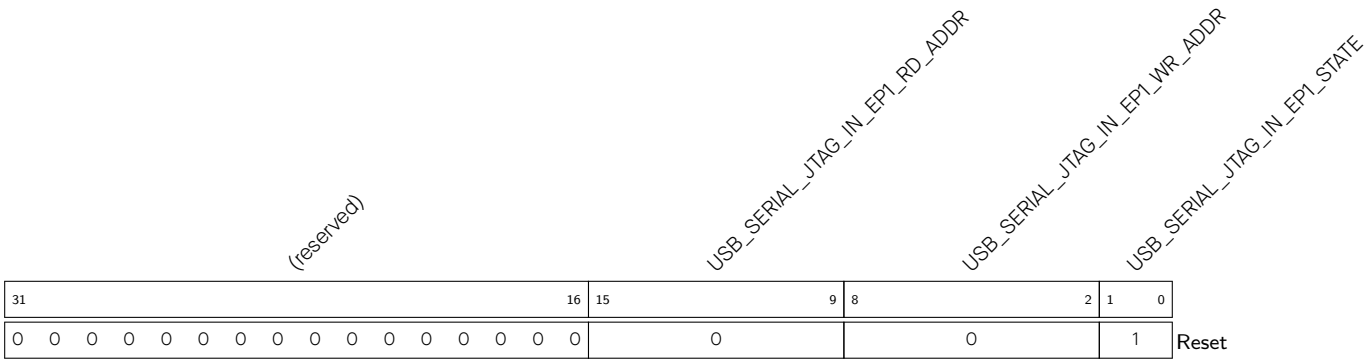


USB_SERIAL_JTAG_IN_EP0_STATE Represents state of IN Endpoint 0. (RO)

USB_SERIAL_JTAG_IN_EP0_WR_ADDR Represents write data address of IN endpoint 0. (RO)

USB_SERIAL_JTAG_IN_EP0_RD_ADDR Represents read data address of IN endpoint 0. (RO)

Register 51.19. USB_SERIAL_JTAG_IN_EP1_ST_REG (0x002C)

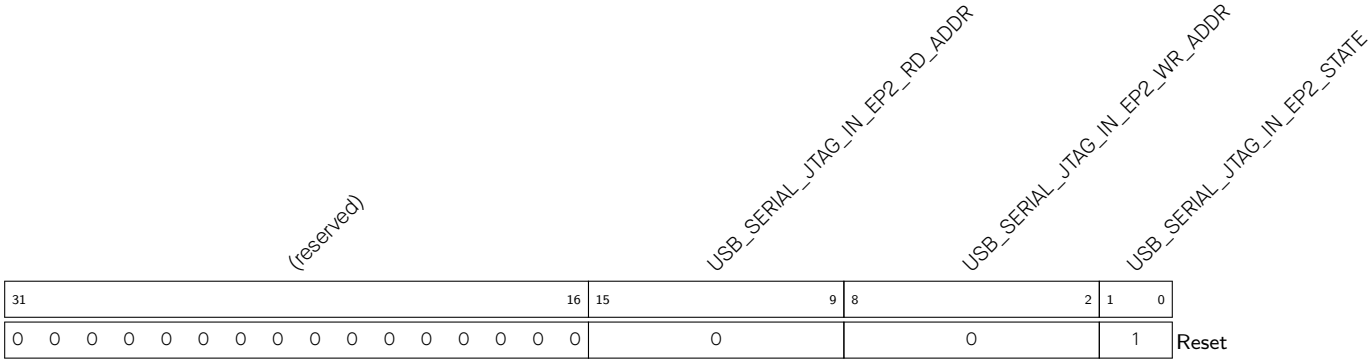


USB_SERIAL_JTAG_IN_EP1_STATE Represents state of IN Endpoint 1. (RO)

USB_SERIAL_JTAG_IN_EP1_WR_ADDR Represents write data address of IN endpoint 1. (RO)

USB_SERIAL_JTAG_IN_EP1_RD_ADDR Represents read data address of IN endpoint 1. (RO)

Register 51.20. USB_SERIAL_JTAG_IN_EP2_ST_REG (0x0030)

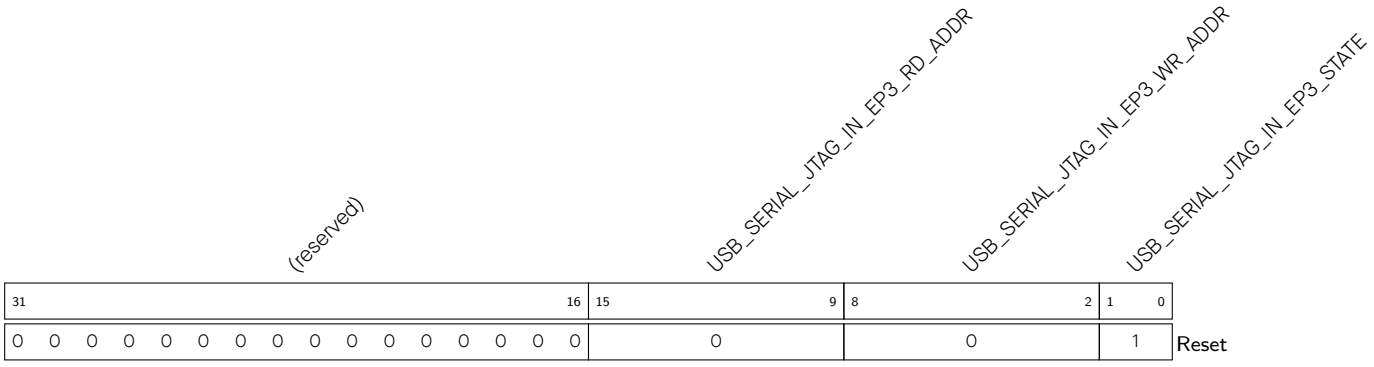


USB_SERIAL_JTAG_IN_EP2_STATE Represents state of IN Endpoint 2. (RO)

USB_SERIAL_JTAG_IN_EP2_WR_ADDR Represents write data address of IN endpoint 2. (RO)

USB_SERIAL_JTAG_IN_EP2_RD_ADDR Represents read data address of IN endpoint 2. (RO)

Register 51.21. USB_SERIAL_JTAG_IN_EP3_ST_REG (0x0034)

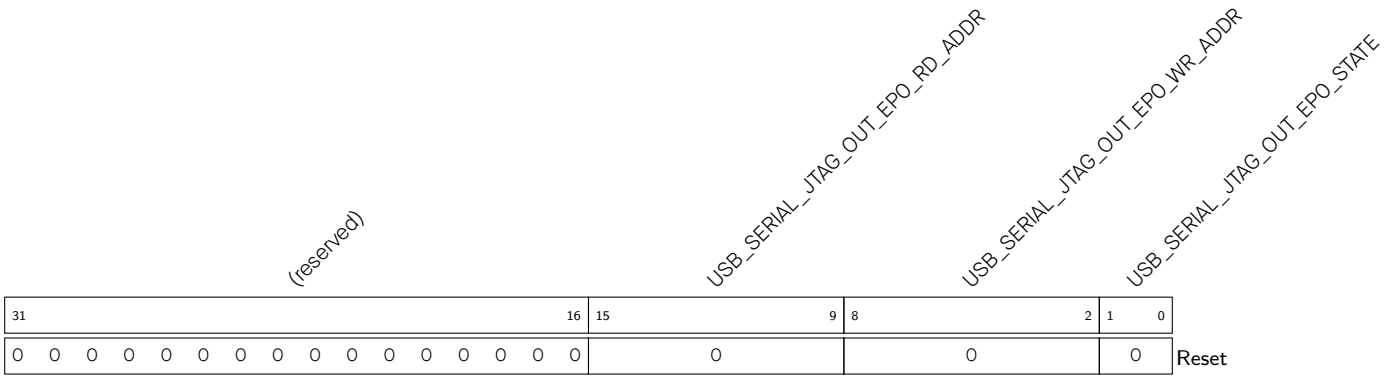


USB_SERIAL_JTAG_IN_EP3_STATE Represents state of IN Endpoint 3. (RO)

USB_SERIAL_JTAG_IN_EP3_WR_ADDR Represents write data address of IN endpoint 3. (RO)

USB_SERIAL_JTAG_IN_EP3_RD_ADDR Represents read data address of IN endpoint 3. (RO)

Register 51.22. USB_SERIAL_JTAG_OUT_EP0_ST_REG (0x0038)



USB_SERIAL_JTAG_OUT_EP0_STATE Represents state of OUT Endpoint 0. (RO)

USB_SERIAL_JTAG_OUT_EP0_WR_ADDR Represents write data address of OUT endpoint 0. When USB_SERIAL_JTAG_SERIAL_OUT_RECV_PKT_INT is detected, there are (USB_SERIAL_JTAG_OUT_EP0_WR_ADDR − 2) bytes data in OUT endpoint 0. (RO)

USB_SERIAL_JTAG_OUT_EP0_RD_ADDR Represents read data address of OUT endpoint 0. (RO)

Register 51.23. USB_SERIAL_JTAG_OUT_EP1_ST_REG (0x003C)

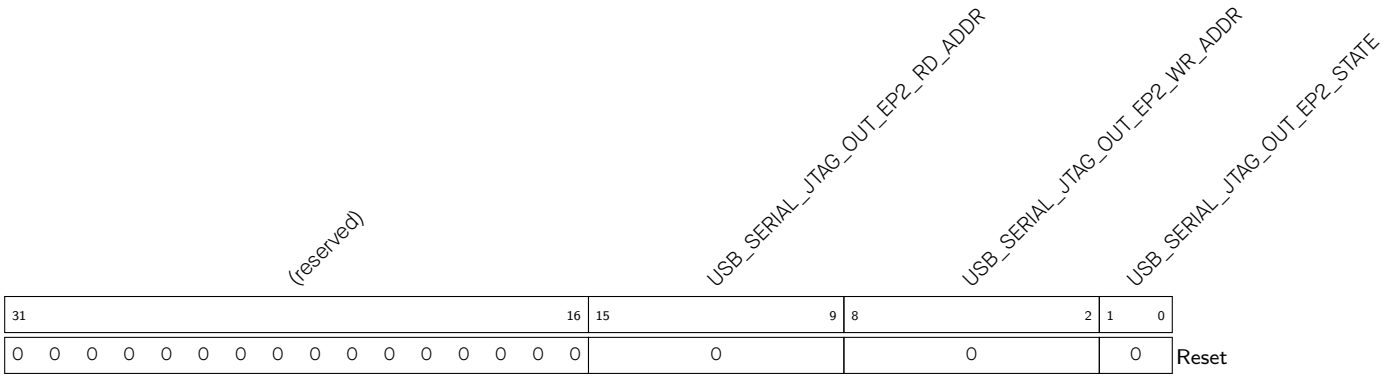
USB_SERIAL_JTAG_OUT_EP1_STATE Represents state of OUT Endpoint 1. (RO)

USB_SERIAL_JTAG_OUT_EP1_WR_ADDR Represents write data address of OUT endpoint 1. When USB_SERIAL_JTAG_SERIAL_OUT_RECV_PKT_INT is detected, there are (USB_SERIAL_JTAG_OUT_EP1_WR_ADDR − 2) bytes data in OUT endpoint 1. (RO)

USB_SERIAL_JTAG_OUT_EP1_RD_ADDR Represents read data address of OUT endpoint 1. (RO)

USB_SERIAL_JTAG_OUT_EP1_REC_DATA_CNT Represents data count in OUT endpoint 1 when one packet is received. (RO)

Register 51.24. USB_SERIAL_JTAG_OUT_EP2_ST_REG (0x0040)

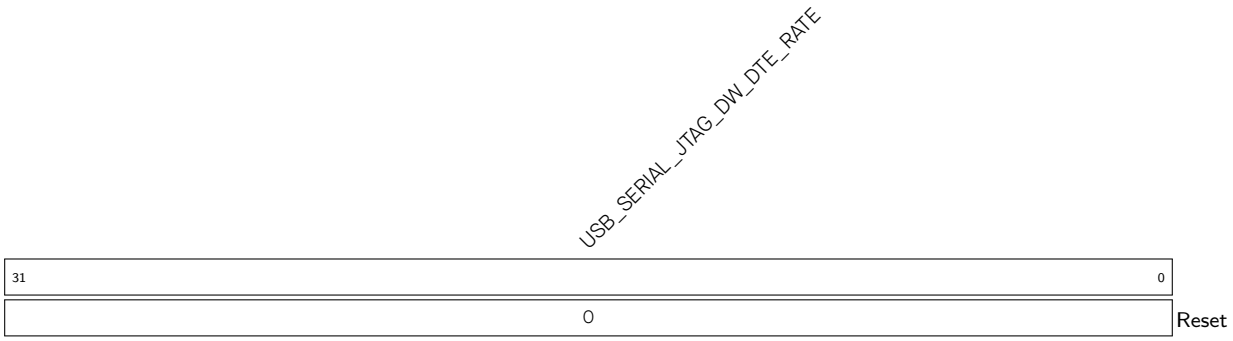


USB_SERIAL_JTAG_OUT_EP2_STATE Represents state of OUT Endpoint 2. (RO)

USB_SERIAL_JTAG_OUT_EP2_WR_ADDR Represents write data address of OUT endpoint 2. When USB_SERIAL_JTAG_SERIAL_OUT_RECV_PKT_INT is detected there are (USB_SERIAL_JTAG_OUT_EP2_WR_ADDR − 2) bytes data in OUT endpoint 2. (RO)

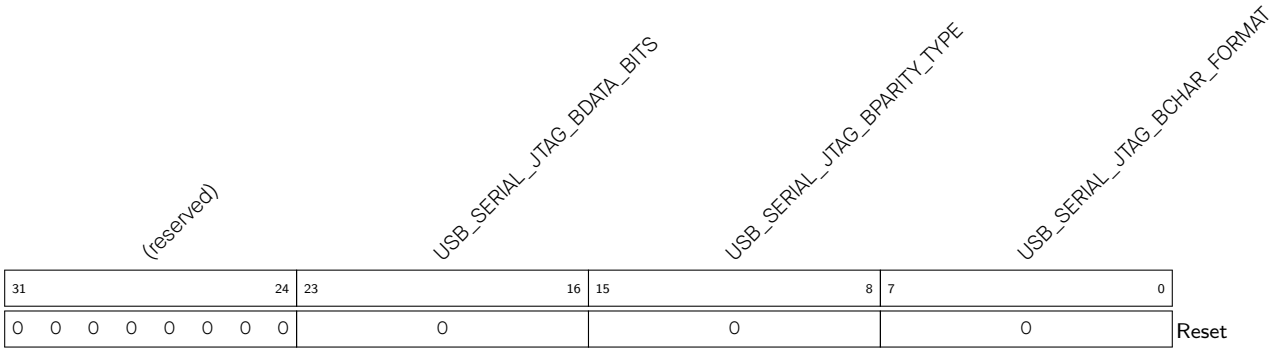
USB_SERIAL_JTAG_OUT_EP2_RD_ADDR Represents read data address of OUT endpoint 2. (RO)

Register 51.25. USB_SERIAL_JTAG_SET_LINE_CODE_W0_REG (0x0050)



USB_SERIAL_JTAG_DW_DTE_RATE Represents the value of dwDTERate set by host through SET_LINE_CODING command. (RO)

Register 51.26. USB_SERIAL_JTAG_SET_LINE_CODE_W1_REG (0x0054)

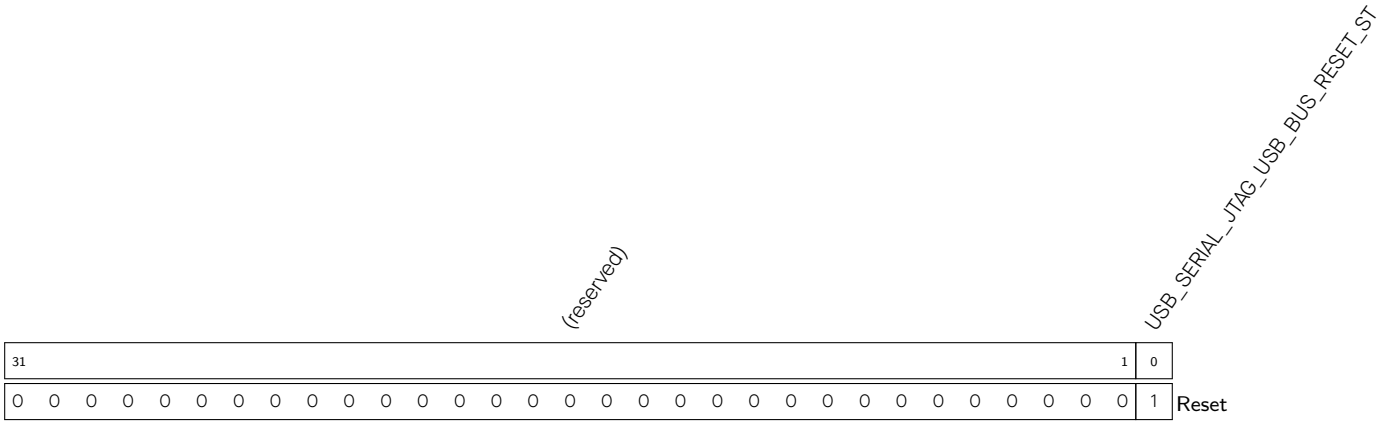


USB_SERIAL_JTAG_BCHAR_FORMAT Represents the value of bCharFormat set by host through SET_LINE_CODING command. (RO)

USB_SERIAL_JTAG_BPARITY_TYPE Represents the value of bParityTpye set by host through SET_LINE_CODING command. (RO)

USB_SERIAL_JTAG_BDATA_BITS Represents the value of bDataBits set by host through SET_LINE_CODING command. (RO)

Register 51.27. USB_SERIAL_JTAG_BUS_RESET_ST_REG (0x0068)



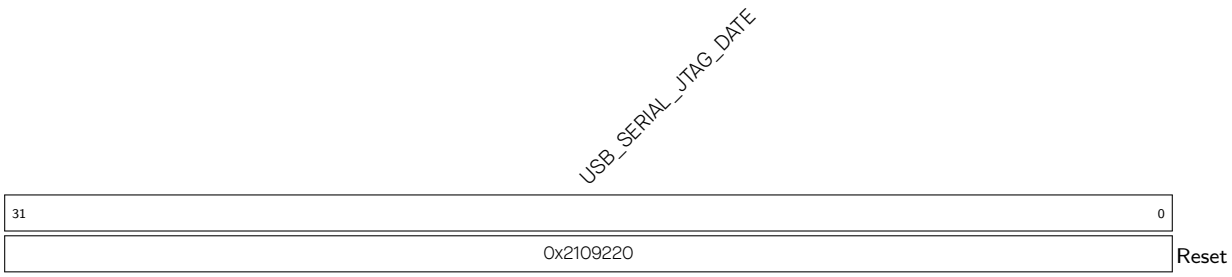
USB_SERIAL_JTAG_USB_BUS_RESET_ST Represents whether USB bus reset is released.

0: USB Serial/JTAG is in USB bus reset status

1: USB bus reset is released

(RO)

Register 51.28. USB_SERIAL_JTAG_DATE_REG (0x0088)



USB_SERIAL_JTAG_DATE Version control register. (R/W)